Bauhaus-Universität Weimar

JOHNS HOPKINS UNIVERSITY

From FEM to Neural Operators: Scientific Machine Learning for Computational Mechanics

Assembly & Boundary Conditions

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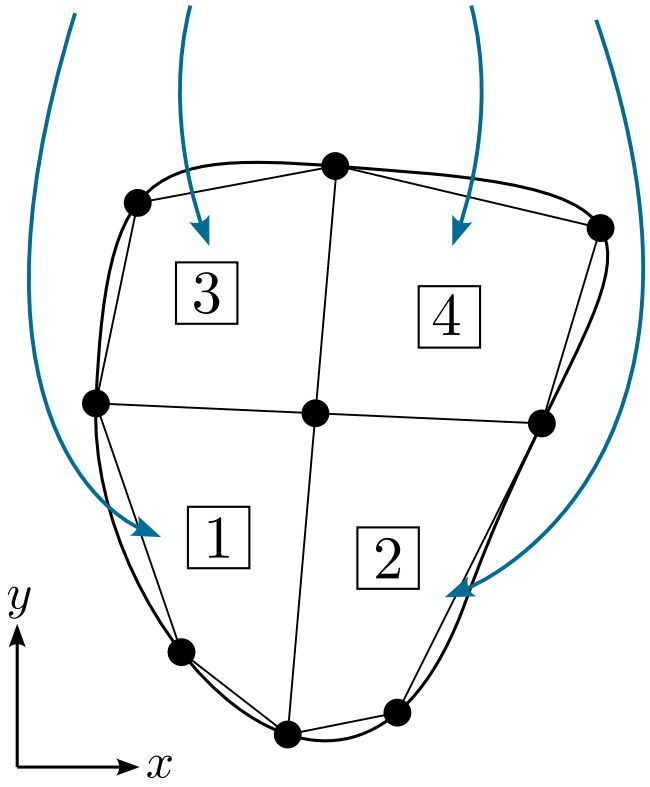
Roadmap

$$K_{ab}^{(e)} = \int_{\Omega_e} \nabla N_a \cdot \nabla N_b \, d\Omega$$

$$f_a^{(e)} = \int_{\Omega_e} f \, N_a \, d\Omega$$



$\Longrightarrow$

Assembly:

$$\mathbf{K}\mathbf{u} = \mathbf{f}$$

$$\Downarrow$$

Boundary Conditions:

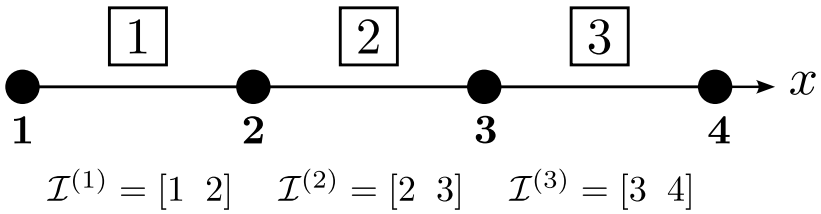
$$\mathbf{K}\mathbf{u} = \mathbf{f}$$

$$\Downarrow$$

Solution:

$$\mathbf{u} = \mathbf{K}^{-1}\mathbf{f}$$

Element Connectivity and Assembly Operator

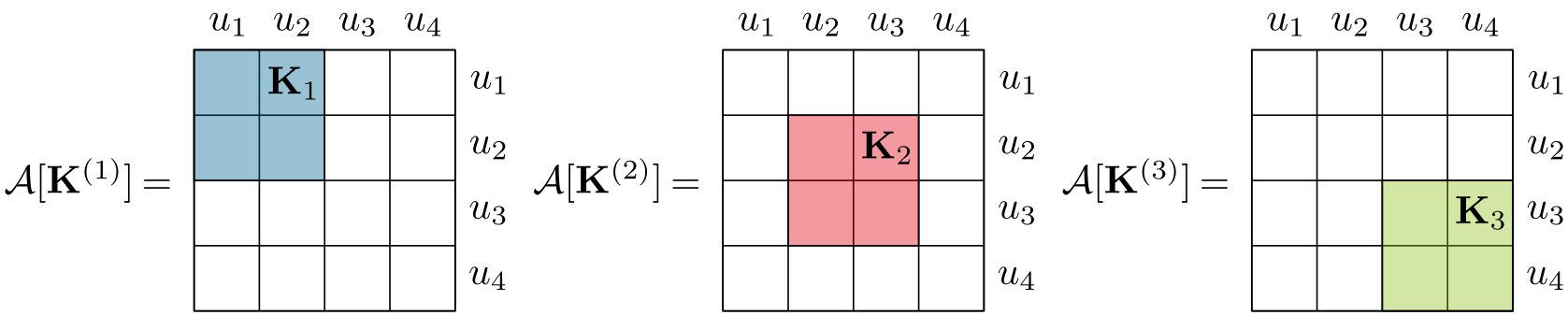


- **Element Connectivity Index:** $\mathcal{I}^{(e)} = (I_1^{(e)}, \ldots, I_{n_e}^{(e)})$
  $\longrightarrow$ Global DOFs corresponding to element e
- **Assembly Operator:**

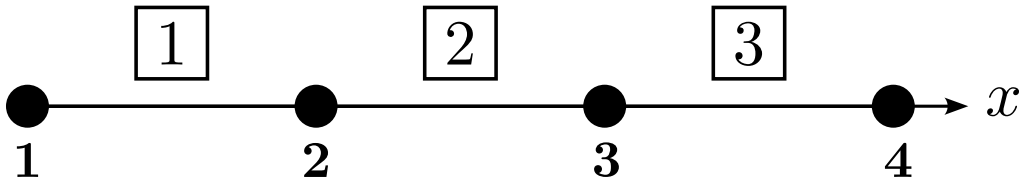
$$\mathcal{A}_e[\mathbf{K}^{(e)}]_{ij} := \sum_{a=1}^{n_e} \sum_{b=1}^{n_e} \left[\, i = \mathcal{I}_a^{(e)} \,\right] \left[\, j = \mathcal{I}_b^{(e)} \,\right] K_{ab}^{(e)}$$

where $[\cdot] \begin{cases} 1 \text{ if true} \\ 0 \text{ otherwise} \end{cases}$

$$\mathcal{A}_e[\mathbf{f}^{(e)}]_i := \sum_{a=1}^{n_e} \left[\, i = \mathcal{I}_a^{(e)} \,\right] f_a^{(e)}$$



Building the global system (1D)



$$\mathcal{I}^{(1)} = [1 \;\; 2] \quad \mathcal{I}^{(2)} = [2 \;\; 3] \quad \mathcal{I}^{(3)} = [3 \;\; 4]$$

- Add the element contributions into the global system::

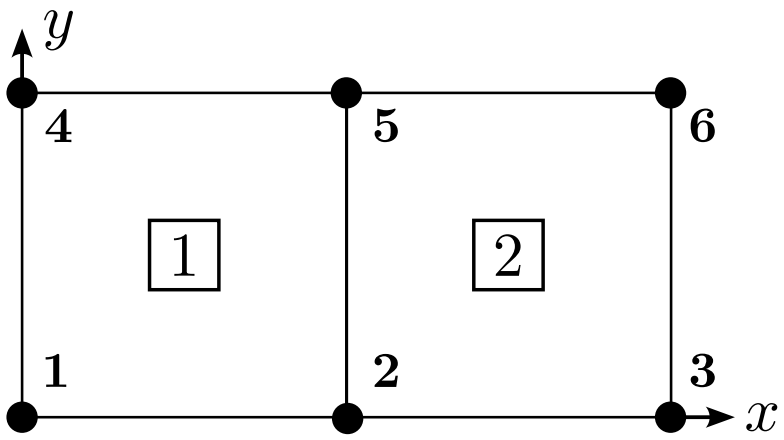
$$K_{ij} = \sum_e \mathcal{A}_e [\mathbf{K}^{(e)}]_{ij},$$

$$f_i = \sum_e \mathcal{A}_e [\mathbf{f}^{(e)}]_i.$$

- Same pattern for any element type (any order p, any number of local DOFs).

$$\mathbf{K} = \begin{array}{c|cccc} & u_1 & u_2 & u_3 & u_4 \\ \hline u_1 & & \mathbf{K}_1 & & \\ u_2 & & & \mathbf{K}_2 & \\ u_3 & & & & \mathbf{K}_3 \\ u_4 & & & & \end{array}$$

Assembly for 2D-Problems

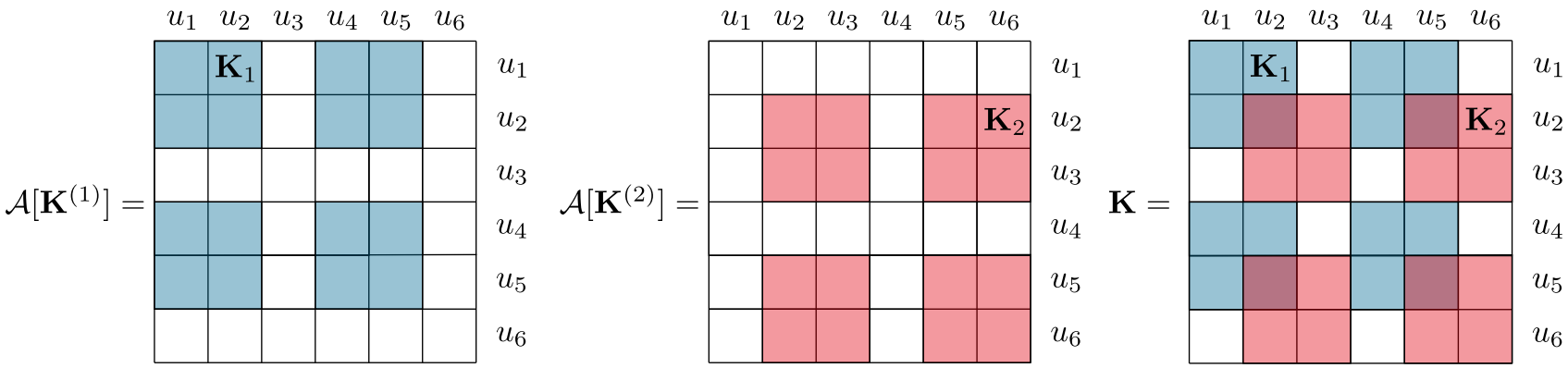


$\mathcal{I}^{(1)} = [1\ 2\ 4\ 5] \quad \mathcal{I}^{(2)} = [2\ 3\ 5\ 6]$

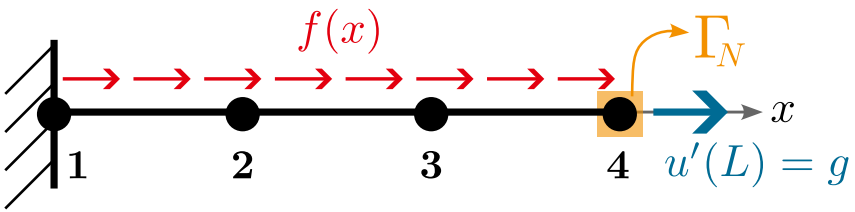
Same idea:

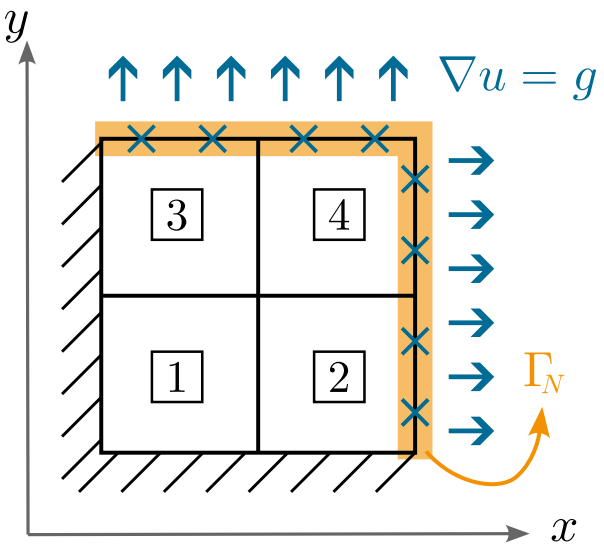
$$K_{ij} = \sum_e \mathcal{A}_e[\mathbf{K}^{(e)}]_{ij},$$

$$f_i = \sum_e \mathcal{A}_e[\mathbf{f}^{(e)}]_i.$$



Neumann (natural) boundary conditions





1D bar (Neumann at $x = L$):

$$\int_{\Gamma_N} g\, v\, d\Gamma \;\Rightarrow\; g\, v(L),$$

only the basis function of the node at L is nonzero, so

$$f_{i_L}^{(\Gamma)} \mathrel{+}= g(L)\, N_{i_L}(L) = g(L).$$

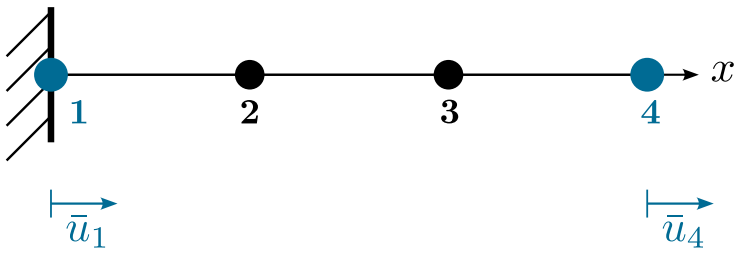
2D (distributed flux on an edge Γ_N):

$$t_a^{(e,\Gamma)} \approx \sum_q w_q\, g(\mathbf{x}_q)\, N_a(\xi_q, \eta_q)\, J_\Gamma(\xi_q, \eta_q),$$

with the *edge* Jacobian

$$J_\Gamma(\xi_q, \eta_q) = \left\| \frac{\partial \mathbf{x}}{\partial s}(\xi_q, \eta_q) \right\|,$$

Dirichlet (essential) boundary conditions



General idea

Split unknowns into free and prescribed DOFs:

$$\mathbf{U} = \begin{bmatrix} \mathbf{U}_f \\ \mathbf{U}_d \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fd} \\ \mathbf{K}_{df} & \mathbf{K}_{dd} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_f \\ \mathbf{f}_d \end{bmatrix}.$$

The global system becomes:

$$\mathbf{K}_{ff}\,\mathbf{U}_f = \mathbf{f}_f - \mathbf{K}_{fd}\,\bar{\mathbf{U}}_d.$$

Example:

Original system: $\mathbf{U} = \begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \\ \bar{u}_4 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{21} & K_{22} & K_{23} & K_{24} \\ K_{31} & K_{32} & K_{33} & K_{34} \\ K_{41} & K_{42} & K_{43} & K_{44} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}.$

Reduced system: $$\underbrace{\begin{bmatrix} K_{22} & K_{23} \\ K_{32} & K_{33} \end{bmatrix}}_{\mathbf{K}_{ff}} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} f_2 \\ f_3 \end{bmatrix}}_{\mathbf{f}_f} - \underbrace{\begin{bmatrix} K_{21} & K_{24} \\ K_{31} & K_{34} \end{bmatrix}}_{\mathbf{K}_{fd}} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_4 \end{bmatrix}.$$

Wrap-up: Practical Notes and Outlook

Practical checklist

- Consistent element connectivity and boundary edge orientation.
- Correct use of the Jacobians for the volume and boundary integrals.
- Make sure outward normals are consistent on Neumann boundaries.
- After applying Dirichlet BCs, solve only for the free DOFs.

Outlook: link to machine learning for scientific computations

- In PINNs, PDE residuals and boundary conditions can appear directly in the loss function.
- Neural Operators are often trained on data generated by FEM solvers assembled in this way.